Universal Skill Expansion Frontiers

INVENTORY GAP ANALYSIS & STRATEGIC ACQUISITION BLUEPRINT

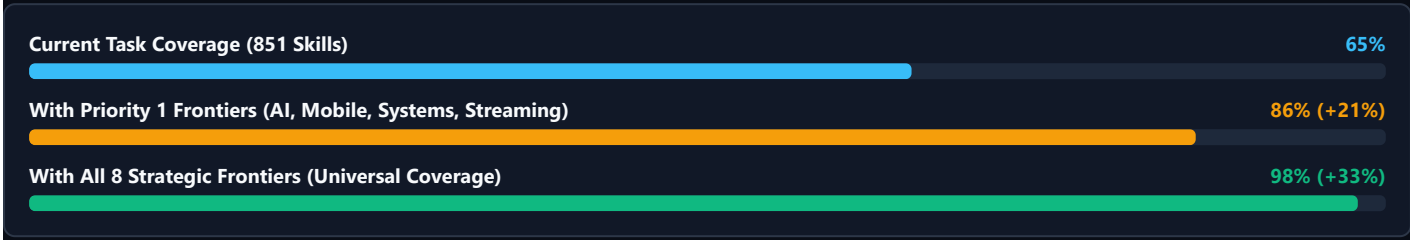
TARGET: 40 CORE SKILLS

851	8	40	98%
CURRENT SKILL TOTAL	STRATEGIC FRONTIERS	TARGET NEW SKILLS	PROJECTED COVERAGE
Across 23 Categories	Critical Growth Domains	Full Stack Coverage	From ~65% current baseline

CURRENT DOMAIN COVERAGE VS IDENTIFIED EXPANSION FRONTIERS

ENGINEERING DOMAIN	CURRENT DEPTH	STATUS ASSESSMENT	TARGET EXPANSION PRIORITY
Python Backend & APIs	143 skills	EXCELLENT (95%)	Maintain & focus on high-throughput async
Frontend & Web UI	79 skills	STRONG (85%)	Add Next.js 15, Turbopack, advanced state
Local AI & LLMOps	8 skills	CRITICAL GAP (25%)	**P0: Unsloth, vLLM, LangGraph, pgvector**
Mobile & Cross-Platform	3 skills	CRITICAL GAP (15%)	**P0: Flutter, React Native Expo, SwiftUI**
Systems & Low-Level	13 skills	CRITICAL GAP (20%)	**P0: Rust Systems, C++23, WebAssembly**
Streaming & Embedded OLAP	4 skills	CRITICAL GAP (15%)	**P0: Kafka, DuckDB, Polars, ClickHouse**
Enterprise Backend	6 skills	HIGH GAP (20%)	**P1: Spring Boot 3 (Java 21), .NET 9 (C# 13)**
Desktop Applications	2 skills	CRITICAL GAP (10%)	**P1: Tauri 2.0 (Rust), PySide6 / Qt**
3D, WebGPU & Shaders	5 skills	HIGH GAP (15%)	**P1: Three.js, WebGPU WGSL, Blender Python**
DevSecOps & Reliability	18 skills	MODERATE GAP (45%)	**P2: Falco Runtime, Chaos Mesh, k6 load**

ANTIGRAVITY TASK COVERAGE EVOLUTION



Expansion Frontiers 1 – 4: Core Engineering Gaps

LOCAL AI INFRASTRUCTURE, MOBILE NATIVE, SYSTEMS PROGRAMMING, REAL-TIME DATA

P0 IMMEDIATE PRIORITY

FRONTIER 1: MODERN AI ENGINEERING, LOCAL INFERENCE & FINE-TUNING

PRI	TARGET SKILL	PRIMARY STACK	TARGET CAPABILITY
P0	**llm-finetuning-unsloth-lora**	Unsloth, PEFT, Axolotl, LoRA	Fine-tune Llama 3 / Mistral on local GPUs with minimal VRAM
P0	**vllm-inference-serving**	vLLM, TensorRT-LLM, SGLang	High-throughput local LLM server with continuous batching
P1	**langgraph-agent-workflows**	LangGraph, StateGraph	Cyclical stateful multi-agent systems with human approval
P1	**vector-db-pgvector-qdrant**	pgvector, Qdrant, Milvus	HNSW vector indexing and hybrid sparse-dense filtering

FRONTIER 2: MOBILE & CROSS-PLATFORM NATIVE ENGINEERING

PRI	TARGET SKILL	PRIMARY STACK	TARGET CAPABILITY
P0	**flutter-dart-architecture**	Flutter 3, Dart, Riverpod, BLoC	Cross-platform native iOS & Android applications
P0	**react-native-expo**	React Native, Expo, NativeWind	Mobile apps using React and TypeScript with native bridges
P1	**swiftui-ios-architecture**	SwiftUI, Swift 6, SwiftData	Native iOS, macOS & visionOS UI and structured concurrency
P1	**kotlin-multiplatform-kmp**	KMP, Compose Multiplatform	Shared business logic and networking across Android & iOS

FRONTIER 3: SYSTEMS & LOW-LEVEL

PRI	TARGET SKILL	STACK
P0	**rust-systems-engineering**	Rust 2024, Tokio, Axum
P1	**cpp-modern-architecture**	C++20/23, CMake, Clang
P1	**webassembly-wasm-wasi**	Wasmtime, Rust WASM
P2	**ebpf-linux-observability**	eBPF, Cilium, bpftrace

FRONTIER 4: REAL-TIME & OLAP

PRI	TARGET SKILL	STACK
P0	**apache-kafka-streaming**	Kafka, Redpanda, Schema
P0	**duckdb-embedded-analytics**	DuckDB, Parquet, PyIceberg
P1	**polars-fast-dataframes**	Polars (Rust/Python)
P1	**clickhouse-olap**	ClickHouse, MergeTree

Expansion Frontiers 5 – 8 & Acquisition Pipeline

ENTERPRISE BACKENDS, DESKTOP RUNTIMES, 3D SPATIAL COMPUTING, DEVSECOPS

P1 & P2 STRATEGIC PRIORITY

FRONTIER 5: ENTERPRISE BACKENDS

PRI	TARGET SKILL	STACK
P0	**spring-boot-java-enterprise**	Spring Boot 3, Java 21
P0	**dotnet-csharp-architecture**	.NET 9, C# 13, EF Core
P1	**go-microservices-chi-grpc**	Go, gRPC, SQLC
P1	**graphql-federation-apollo**	Apollo Federation v2

FRONTIER 6: DESKTOP APPLICATIONS

PRI	TARGET SKILL	STACK
P0	**tauri-desktop-development**	Tauri 2.0 (Rust + Web)
P1	**electron-app-security**	Electron 30+, ContextBridge
P1	**pyside6-qt-applications**	PySide6, Qt 6, QML

FRONTIER 7: 3D, WEBGPU & SHADERS

PRI	TARGET SKILL	STACK
P0	**threejs-webgl-experiences**	Three.js, React Three Fiber
P1	**webgpu-compute-shaders**	WebGPU, WGSL shaders
P1	**blender-python-automation**	Blender bpy API

FRONTIER 8: DEVSECOPS & CHAOS

PRI	TARGET SKILL	STACK
P0	**kubernetes-runtime-security**	Falco, Tetragon, OPA
P1	**chaos-engineering-resiliency**	Chaos Mesh, Toxiproxy
P1	**k6-distributed-load-testing**	Grafana k6 (JS / Go)

AUTOMATED ACQUISITION & KNOWLEDGE FABRIC INTEGRATION

1. COMPILE
Generate SKILL.md specs from authoritative docs

2. AUDIT
Run SkillValidator & path guard checks

3. INGEST
Rebuild Knowledge Fabric graph nodes & CTEs

4. ACTIVATE
Instant multi-mode RRF retrieval in Antigravity